Secure Data Integrity and Anomaly Detection Framework for Financial Systems

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***Abstract*—Financial transactions consist significant confidential data, are highly regulated, and therefore these systems need to be guaranteed with integrity and good detection of any deviated behavior. The present literature review focuses on two related domains: the secure end-to-end data-processing flow to provide correctness, traceability, and tamper resistance, and anomaly-detecting machine-learning-based models, to detect and differentiate fraudulent and synthetically injected behavioral anomalies. Existing studies focus on huge risks of validity posed by sophisticated cyber-threats, complexity of infrastructures, and AI-based systems. Similar research proves the use of supervised, hybrid, and deep learning as effective solutions to detect financial fraud; however, the research highlights issues with data imbalance, explainability, and applicability in a real-time setting. However, there is still a literature gap, lack of integrity assurance and ML-based detection framework mechanisms that have not been properly integrated to achieve a single secured entity. The review identifies critical gaps and sets the standard for an integral secure software engineering strategy that would utilize consistency-confirmed data pipelines together with malicious anomaly detection in financial operations and environments.**

I. Introduction

Financial systems mainly deal with the processes that are conducted in a very interconnected and technology-intensive settings where enormous amount of sensitive transactional data is operated on a consistent basis. The growth rate of the fintech systems, digital banking systems, and artificial intelligence-driven systems has dramatically altered the conventional financial activities [1]. Information reliability has become a primary security issue in this domain. Most of the Finance-based organizations are susceptible to the advanced persistent threats (APTs) and infrastructural vulnerability that may produce a risk to transactions data [2].

These concerns are enhanced by the fact that AI and machine learning have been integrated into financial management systems. Breach of data inputs may have a direct impact on the automated decision-making and compliance monitoring systems and credit scoring [3] [4]. Additionally, more incorporation of data between the heterogeneous platforms can give a new paradigm for data analysis on a regular basis [5] but does not guarantee data protection.

Equivalent to the integrity issue, machine learning has gained prominence in terms of fraud detection and identification of behavioral anomalies. Performance in detection has become more effective through supervised and ensemble learning techniques than conventional rule-based methods [6] [7]. Deep learning and hybrid models offer even stronger detection of complex patterns of transactions [8] [9]. Nevertheless, there are still severe obstacles, especially in the field of data disproportion, explainability, and inability to deploy in real-time [10] [11].

Despite several research studies conducted on cybersecurity controls and ML-based fraud detection, there is no single framework that considers both in an environment of safe transactions. The existing literature lacks evaluation of the impact of integrity assurance mechanisms on model performance and does not examine an experiment on detection strength in controlled settings where ML datasets are corrupted with behavioral anomalies. Subsequently, it is not certain whether anomaly-detecting systems are reliably deployed in tamper-resistant and traceable financial data streams. The aim of this study is to address the need of a secure environment or framework consisting of integrity-checks at each step for monitoring attributes like tamper resistance, traceability, and validation consistency, to assess any deviation under such conditions, and the necessity of data assurance for testing the potential of fraud detection when using ML models. In order to evaluate this, the literature focuses on the research in financial systems as-

- To design a secure end-to-end data-processing framework that ensures data trustworthiness, traceability, and tamper resistance.
- To assess and determine synthetically injected anomalies in the systems when using machine-learning-based models to achieve valid and actual results.

II. Related Work

The existing literature addresses threats, like sophisticated persistent threats, that arise out of governance, and threats caused by infrastructures. Meanwhile, the literature of ML is concerned with the performance of anomaly detection, usually with the assumption that input data is safe and reliable.

A. End-to-End Data Integrity and Secure Framework Design

1) Cyber Threats and Organisational Readiness: Zainudin et al. [2] focus on financial institutional (FI) preparedness against advanced persistent threats (APT), enlightens limitations in monitoring systems, incident response mechanisms,

and the strategic plan of detecting long-term threats. Conversely, Zamil et al. [12] consider a wider and review-oriented approach to investigate the issues of cybersecurity and data completeness within the FI based on their governance frameworks, compliance systems, and systemic risk reduction. The difference in both the emphases is organisational preparedness and systemic governance, which shows that the risk of accuracy arises both as a matter of immature operations and as a result of constraints within operational structure.

2) Infrastructure Complexity and Data Protection Challenges: Systematically reviewing the cybersecurity threats of fintech innovation, Ogunleye et al. [13] emphasize the progress of technological adoption that has enhanced exposure to unlicensed access, monetary loss, phishing attacks and compliance issues in interconnected banking ecosystems. Similarly, Kumar et al. [14] issues data security practices directly to consumer trust that considers insufficient protection systems, which impact reliability and institutional trust in digital financial platforms. First study dwells upon structural cybersecurity risk that arise due to fintech innovation and ecosystem complexity, whereas other dwells upon the trust implications that weak data protection carries along with it.

3) Emerging Technologies and Data Integration Risks: Chien et al. [5] discuss financial data integration by visualization using knowledge graphs and show that aggregation of large amounts of data enhances visibility and decision-making, but leads to the spread of corrupted or inconsistent information when there is a weak validity verification tool. Duggal et al. [3] examine how AI and ML affect financial management systems and state that sophisticated analytics increase reliance on viable and dependable data processes. Deshpande et al. [4] also address the problem of AI-related cybersecurity threats, pointing out that it is possible to directly attack automated financial decision systems by manipulating training or transactional data in a malicious way. Primary study concentrates on the risks within the integration layer, whereas both the other investigations highlight the vulnerability caused by AI and finding that technological progress is making transaction information more dependent and exposed.

4) Technical and Architectural Integrity Controls: Nadji et al. [15] state that the concept of architectural safeguards, including hashing, encryption, and access validation, can be the cornerstone of securing the transactional records against criminals. Expanding on this architectural view, Kashyap [16] conducts a review of Zero Trust Architecture as a contemporary model of enforcement that requires unremitting identity validation, severe access segmentation and dynamic validation of components in the system. First research focuses on cryptography enforcement controls at data level, while the other, enlightens with a paradigm of continuous verification at architecture level, as typical examples of complementary but different means to enhance structural network security.

B. Machine Learning–Based Behavioural Anomaly Detection

1) Supervised ML Techniques: Compagnino et al. [6] discuss the use of machine learning to detect fraud under supervised learning algorithms including logistic regression, decision trees, random forests, and support vector machine, and noting that they are generally effective when trained on labeled financial data. On the same note, Martinez et al. [7] perform a comparative empirical analysis of several ML models used to identify fraud banking transactions and show that ensemble and advanced techniques are more effective in terms of detection and recall than single baseline models. First study concentrates on the methodological aspects and the suitability of different algorithms, but the next study pays much attention to the performance benchmarking and the comparison of models.

2) Hybrid and Deep Learning Approaches: Shanaa et al. [8] suggest a hybrid framework of anomaly detection, which incorporates both supervised and unsupervised learning methods to enhance the process of detecting fraud, especially in cases where the patterns of transactions are uncommon and unknown. Conversely, Zerine et al. [9] concentrate on AI-based and deep learning methods of fraud detection and their capacity to identify various complex behavioural and temporal patterns among series of transactions. Initial study focuses on the balance between structured detection and anomaly discovery, while the second looks at the tradeoff between complexity modeling and pattern depth, which has a higher computational cost and lower interpretability.

3) System Integration and Real-Time Detection: Basani et al. [11] discuss data engineering pipeline and machine learning pipeline integration to detecting impossible fraud in real-time processes with the focus on how efficient the data ingestion, the feature engineering, and low-latency processing capabilities are to detect all the financial transactions. Comparatively, Hernandez Aros et al. [10] conduct a thorough literature review of various machine learning methods in financial fraud detection and detects some common problems (imbalance in data, scalability, metrics of evaluation, and deployment limitations). When first research centers on a viable system-level integration to realize real-time detection, the next one creates wider methodological and performance-related constraints of currently existing ML approaches.

The overall outcome of the reviews shows that financial environments operate across interconnected risks at organisational, infrastructural, and algorithmic levels. On the organisational level, the advanced persistent threats (APTs), phishing attacks, malware infections, data breaches, abuse of insider privileges or insider manipulation, and infiltration of the system over time provide possibilities of unidentified manipulation of transaction records. On infrastructural level distributed architectures, third party dependencies, API gateways, and heterogeneous data flows increase attack surface and increase exposure to data tampering during ingestion, transmission or storage. Algorithmic controls include cryptographic hashing, encryption, access checks, and privacy-preserving computation, suggested to control these risks, but these controls are usually considered in isolation rather than evaluating them in a holistic transaction-processing system. At the same time, machine-learning-based fraud-detection-studies

indicate a better predictive accuracy, recall, and convergence efficiency with combined, hybrid and deep learning methods but the assessment is done on benchmark datasets that assume structurally reliable and trustworthy input data.

This leaves a number of unaddressed issues: none of it can be tested on validity of correctness across transaction stages, none of it is tested as robust in a behaviorally adversarial environment, there is little understanding of how architectural protection effects on model performance, and limits to evidence of the stability of the detection engine in secured deployment. With an ever-increasing dependency on AI-based analytics in financial systems coupled with the operation in a threat-intensive environment, the relationship between structural protection mechanisms and behavioural anomaly detection is an area of critical but under-studied concern. Research is hence required by the lack of verifiable evaluation frameworks that at the same time address the transaction immutability, enforcement of traceability, consistency of validation and machine-learning detection performance under controlled adversarial cases.

In both domains, it can be considered that there is no architectural unification of integrity assurance mechanisms and anomaly detection models. To assess these findings, the table below summarizes the key integrity risks and ML detection limitations identified across the existing studies:

TABLE I
DATA INTEGRITY RISKS AND ML DETECTION LIMITATIONS IN FINANCIAL SYSTEMS

Study	**Focus Area**	**Key Risk / Contribution**	**Identified Limitation**
Zainudin et al. [2]	APT Readiness	Organisational gaps in detecting advanced persistent threats	No end-to-end integrity pipeline validation
Zamil et al. [12]	Cybersecurity Governance	Data tampering and compliance risks	Lacks empirical architectural implementation
Miguel et al. [17]	Infrastructure Security	Distributed systems increase attack surface	No unified secure integration framework
Nadji et al. [15]	Cryptographic Controls	Hashing, encryption, and access control for integrity assurance	Not evaluated within full transactional process
Compagnino et al. [6]	Supervised ML	Strong performance in labelled fraud detection	Assumes integrity-assured datasets
Shanaa et al. [8]	Hybrid ML	Improved novelty detection using hybrid models	No testing under integrity-verified approach
Basani et al. [11]	Real-Time ML Integration	Importance of data engineering in detection performance	Integrity mechanisms not integrated with ML workflow

III. CONCLUSION

Existing literature demonstrates that the security of the financial systems has been significantly progressed in two key aspects as far as architectural security enforcement and machine-learning-based fraud detection are concerned. Research into organizations and governance has enhanced knowledge on the category of threats that are currently prevalent across the cyberspace and breaches of compliance and technical studies have offered cryptographic protection mechanisms; hashing, encryption, access verification and privacy-preserving computation to safeguard transactional data in both storage and communication. At the same time, machine-learning studies have also met serious progress, labeled, accumulated, hybrid, and deep learning models are more powerful in terms of their detection potential, recall, and behavioural pattern recognition abilities in cases of fraud detection.

In spite of these advancements, the literature indicates a structural detachment between protection mechanism of the system level, and anomaly detection assessment. Research on security is dedicated to unauthorized modification prevention, record consistency, and access control policy enforcement, but it does not factually study the mechanisms of detection models in settings with the active use of such controls. On the contrary, research on anomaly detection focuses on predictive performance, convergence behavior and benchmarking based on data, with a common belief that the input transactions are structurally reliable and uninterrupted. This lack of evaluation integration becomes particularly limiting as the financial platforms become more and more dependent on automated analytics, real-time processing, and decision-making models, as they become deployed in safer (well-secured and traceable) operational conditions.

The current research will help to resolve that weakness, as it will unify the enforcement mechanisms in architecture with the organized anomaly detection testing into a common experimental model. The paper proposes a safeguarding architecture of the transaction-processing in which record immutability is asserted using cryptographic hashing, validation integrity is ensured using controlled verification processes, and the traceability is ensured between processing phases. In this setting, when a deviant behaviour is introduced synthetically i.e. deviation in transaction patterns, behavioural drift and controlled irregularity without undermining structural validation characteristics, the validation result fails, thus, it will not only enhance the security from outside but from inside as well. The hybrid and supervised and deep learning models are then tested not just to be predictive but against robustness, stability and reliability of detection under these secured conditions of deployment. This study adds up to the current state-of-the-art by practically assessing ML robustness within protected transaction architectures, which has not been systematically considered in prior existing work. The result is a framework that comprises of more resilient financial systems- systems that are capable of maintaining structural correctness and at the same time maintaining a reliable behavioural anomaly detection under adversarial and high-dependence environments. The integration of the same is necessary in the future financial infrastructures where the security controls and automated intelligence should not act in isolation of each other but collaboratively.

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